

2.5 Curvature Worksheet

1. Which of the following expressions is equal to the curvature of a curve given by the parameterization $\mathbf{r}(t)$ where $t \geq 0$, $\mathbf{T}(t)$ is the unit tangent vector, and $s(t) = \int_0^t \|\mathbf{r}'(u)\| du$? Simply circle your answer.

You do not need to provide justification.

A. $\left\| \frac{d\mathbf{T}}{dt} \right\|$ B. $\frac{d\mathbf{T}}{ds}$ C. $\frac{d\mathbf{T}}{dt}$ D. $\left\| \frac{d\mathbf{T}}{ds} \right\|$

2. Consider the curve $\mathbf{r}(t) = \langle t^2, \ln(t), t \ln(t) \rangle$. Find the curvature of $\mathbf{r}(t)$ at the point $(1, 0, 0)$.

3. Consider the curve $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$.

a) Determine a formula to measure the curvature of $\mathbf{r}(t)$ at any value of t .

b) Determine the curvature of $\mathbf{r}(t)$ at the point $(1, 0, 2\pi)$, $(1, 0, 0)$, and $(-1, 0, \pi)$.

4. Consider the curve $\mathbf{r}(t) = \langle t^2, 0, t \rangle$.

a) Determine a formula to measure the curvature of $\mathbf{r}(t)$ at any value of t .

b) Determine the curvature of $\mathbf{r}(t)$ at the points $(0,0,0)$, $(1,0,1)$, and $(4,0,2)$.